

# GREEN ALIBI

## Testing Whether Fluorescence Catches Drought Stress Before the Eye Can See It

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### The Question

India's official drought-declaration process, and the PMFBY crop-insurance payout mechanism that depends on it, rely substantially on NDVI — a reflectance signal that only changes once a plant's structure has already begun to visibly degrade. Solar-Induced Fluorescence (SIF) is grounded in a more direct physical process: a drop in photosynthetic efficiency, before outward greenness changes. Does that physical head-start actually show up as a measurable time lag on real satellite data — and does drought severity make that lag bigger, as hypothesized?

### The Method

Eight districts of Marathwada, Maharashtra were tested across eight growing seasons (2015–2023, excluding 2021), using GOSIF v2 fluorescence data and cloud-screened MODIS NDVI. The study originally covered three years (2015, 2018, 2020); five more (2016, 2017, 2019, 2022, 2023) were added in a later expansion specifically to fix the original small-sample limitation. The SIF-to-NDVI decline lag was calculated two independent ways — threshold-crossing (linear-interpolation at 5 seasonal decline thresholds) and time-lagged cross-correlation — with 2,000-replicate bootstrap confidence intervals quantifying precision. Rainfall-anomaly validation against a 20-year CHIRPS climatology found only 2 of the 8 years (2015, 2018) actually meet this study's own drought threshold — the other 5 added years all sit within roughly one standard deviation of normal.

### The Finding

SIF's decline precedes NDVI's in 7 of the 8 years — confirmed in direction by an independent cross-correlation method, and by bootstrap resampling never favoring NDVI leading SIF in any single year (100% of replicates, every year). 2018 is a genuine, replicated exception: its threshold-crossing lag is marginally negative, its cross-correlation lag is exactly zero, and only 8.1% of its bootstrap replicates show a positive lag — three independent methods agreeing 2018 breaks the pattern. Drought severity still does not amplify the lag: the two drought years averaged a smaller lag (7.6 days) than the six normal years (15.0 days) — the same direction as the original 3-year study, now confirmed at more than double the sample size.

| Year           | Rainfall Anomaly | Threshold-Crossing Lag | Cross-Corr. Lag |
|----------------|------------------|------------------------|-----------------|
| 2015 (drought) | −21.5%           | 16.3 days              | 4 days          |
| 2016 (normal)  | +9.5%            | 23.8 days              | 2 days          |
| 2017 (normal)  | −1.1%            | 21.6 days              | 27 days         |
| 2018 (drought) | −18.3%           | −1.1 days              | 0 days          |
| 2019 (normal)  | +13.3%           | 17.2 days              | 18 days         |
| 2020 (normal)  | +29.1%           | 20.6 days              | 0 days          |
| 2022 (normal)  | +36.8%           | 5.1 days               | 0 days          |
| 2023 (normal)  | −3.7%            | 4.2 days               | 0 days          |

H1 (SIF leads NDVI) is supported by both methods in most years, with 2018 as an honestly-disclosed exception across all methods. H3 (drought amplifies the lag) is explicitly reported as not supported — a genuine, non-result confirmed at  $n = 8$  years, not just  $n = 3$ .

### Validation & Robustness Checklist

- ✓ Two independent lag-estimation methods (threshold-crossing + cross-correlation)
- ✓ Bootstrap confidence intervals (2,000 replicates per year, all 8 years)
- ✓ Independent rainfall validation (CHIRPS, 20-year climatological baseline)
- ✓ Moran's I spatial-autocorrelation check (effective sample size disclosed, per year)
- ✓ Cross-referenced against the real government drought-declaration date (2018)
- ✓ Sample expanded from 3 to 8 years — a real, executed fix, not just a stated limitation
- ! H3 (drought amplifies lag) — honestly reported as not supported
- ! 2018 is a consistent exception to H1 across every method — reported, not hidden
- ! Exact year-to-year ranking flagged as method-sensitive, not robust

### Honest Limitation

Only 2 of 8 years meet this study's own drought threshold, so the drought-vs-normal comparison remains a 2-versus-6 split, not a balanced one. Every lag value is a point estimate from 14–18 post-peak satellite observations per year — confidence intervals are wide, and all 28 possible pairwise comparisons between years' intervals overlap, so the exact year-to-year ranking is not statistically distinguishable from noise even at 8 years. The SIF-rainfall spatial correlation (Pearson  $r = 0.567$ ,  $p < 0.0001$ ,  $n = 64$  district-years) is real but notably weaker than the  $r = 0.837$  originally reported from the smaller, drought-heavy 3-year sample — and a Moran's I diagnostic finds SIF's own spatial clustering significant in only 4 of 8 years, versus rainfall's, which is significant in all 8.

### Real-World Relevance

Comparing 2018's SIF and NDVI decline onset against Maharashtra's official drought declaration (31 October 2018) shows both satellite indicators leading the official declaration by roughly seven to eight weeks. After a district-boundary naming bug was found and fixed during the sample expansion, NDVI now crosses its threshold about five days before SIF in 2018 specifically — the reverse of the original finding, and consistent with 2018 being this study's one exception throughout. For a rainfall-deficit-prone farming region, the larger, more reliable opportunity is satellite-based monitoring of either kind arriving seven-plus weeks ahead of the current process — with SIF's own edge over NDVI a smaller, year-dependent refinement on top of that, not a guaranteed one.

**GitHub:** [github.com/sakshimaske303-commits/GREEN\\_ALIBI](https://github.com/sakshimaske303-commits/GREEN_ALIBI) | **Live Dashboard:** [greenalibi-bzs2wvod5fflqh7dfe2cf4.streamlit.app](https://greenalibi-bzs2wvod5fflqh7dfe2cf4.streamlit.app) | **Zenodo DOI:** 10.5281/zenodo.21762501

Sakshi D. Maske — Independent Geospatial Researcher